

# Earthquake cycles and evolving fault permeability: a reproducibility study of Zhu et al. (2020)

Computational research report

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## Abstract

Permeability changes couple fault slip to the redistribution of pore pressure. We implement the model of Zhu et al. [1] in C++ and assess which parts of the published calculation can be recovered from its equations, inputs, and archived outputs. Independent evaluation recovers the constitutive curves and steady pressure profile in Figure 1. The archive permits reconstruction of all six main and six supplementary figures, including the migrating pressure pulses and swarm sequence. These archived trajectories are distinguished throughout from new numerical simulations. The new solver passes analytical and numerical verification tests and generates coupled pressure fronts and seismic slip from documented, approximate restart states. On the two finest meshes, the first featured-case earthquake occurs 38–39 days earlier than in the archive, while the two meshes differ by 1.13 days. Temporal refinement changes its onset by 169 s. These calculations recover the principal feedback but do not establish an exact independent reproduction of the published earthquake sequence.

## 1 Scope and sources

Fault healing reduces permeability between earthquakes. If fluid supply from depth continues, pressure increases and effective normal stress decreases. Slip can then increase permeability and permit the pressure perturbation to advance. Zhu et al. [1] showed that this feedback can produce aseismic slip fronts and swarm-like seismicity. The objective here is to recover that calculation while keeping the distinction between a new model solution and an archived solution explicit.

Three sources constrain the exercise: the article and its supplement, the OSF archive [2], and the Scycle branch `Zhu_et_al_2020` [3]. All ten archive files, including the data-description PDF, were downloaded and checked against their OSF SHA-256 digests. Nine files contain simulation fields. The source branch supplies four input files, the stretched depth grid, and plotting examples, but the OSF files do not include the frictional state or shear traction needed for an exact restart. The C++ implementation, initial profiles, diagnostics, figure scripts, and LaTeX source are available in the [project repository](#). Its release includes a standalone report-source bundle and the full newly computed field records; the original archive is retrieved separately through its checksum-verified download manifest.

## 2 Mechanical and hydraulic model

The fault occupies  $y = 0$  in a rectangular antiplane elastic domain  $0 \leq y \leq L_y$ ,  $0 \leq z \leq L_z$ , with depth  $z$  positive downward. Slip is the displacement discontinuity across the fault. The remote boundary moves at  $V_p/2$  on the modeled side; the upper and lower boundaries have zero shear traction. The

| Published figure | Calculation in this report                                                  | Reproducibility status                                                             |
|------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1                | Geometry redrawn; constitutive and steady-flow equations evaluated anew     | Independent constitutive reproduction.                                             |
| 2–5              | Archived slip, velocity, pressure, permeability, and flux replotted         | Archived numerical results recovered; new sequence not thereby validated.          |
| 6                | Advancing fronts extracted from archived velocity fields                    | Qualitative reconstruction with declared threshold and time windows.               |
| S1–S6            | Archived time histories, alternate cycles, and long-healing cases replotted | Archived results recovered with revised plotting conventions.                      |
| New C++ runs     | Coupled evolution from approximate archived states                          | Independent numerical integration, with initialization and resolution limitations. |

**Table 1:** The provenance of each result. Figures in this report are newly generated plots; original published raster figures are retained only as reference material.

elastic displacement satisfies  $\nabla^2 u = 0$ . For a cosine slip mode with vertical wavenumber  $\kappa_j = j\pi/L_z$ , the exact elastic traction operator is

$$\hat{\tau}_j^{\text{qs}} = K_j \hat{d}_j, \quad K_j = -\frac{\mu \kappa_j}{2 \tanh(\kappa_j L_y)}, \quad K_0 = -\frac{\mu}{2L_y}, \quad (1)$$

where  $d = \delta - V_p t$  is the slip deficit, with any initial elastic stress represented separately. Equation (1) retains the finite-domain boundary conditions without solving for off-fault displacement at every step. It is evaluated with discrete cosine transforms on a uniform depth grid.

The quasi-dynamic traction balance and regularized rate-and-state law are

$$\tau^{\text{qs}} - \eta_{\text{rad}} V = (\sigma - p) f(V, \psi), \quad V = \dot{\delta}, \quad (2)$$

$$f(V, \psi) = a \operatorname{asinh} \left[ \frac{V}{2V_0} e^{\psi/a} \right], \quad (3)$$

$$\dot{\psi} = \frac{b}{d_c} \left[ V_0 e^{(f_0 - \psi)/b} - V \right]. \quad (4)$$

Here  $\psi$  is the logarithmic aging state,  $a$  and  $b$  are depth-dependent friction parameters, and  $d_c$  is the state evolution distance. The radiation damping term approximates the energy carried away by elastic waves. The original input files impose a minimum effective stress of 1 MPa in friction; we retain that prescription and identify it explicitly below. The new implementation assumes the fixed slip direction of the investigated sequences,  $V \geq 0$ , and rejects trial steps with reverse slip. It does not clip negative velocities. All archived and accepted new velocities in this study have the assumed sign. Simulating reversal would require retaining the signed aging law and the original implementation’s  $|V|$  in permeability evolution.

Fluid flow is confined to the fault zone. The governing hydraulic equations are

$$n\beta \partial_t p = \partial_z q, \quad q = \frac{k}{\eta} (\partial_z p - \rho g), \quad (5)$$

$$k = k_{\min,2} + (k^* - k_{\min,2}) e^{-(\sigma - p)/\sigma^*}, \quad (6)$$

$$\partial_t k^* = \frac{V}{L} (k_{\max} - k^*) - \frac{k^* - k_{\min}}{T}. \quad (7)$$

Positive  $q$  denotes upward flow. Pressure is zero at the surface and the bottom flux is  $q_0$ . Compression reduces permeability through Equation (6); slip and healing compete through Equation (7). The

distinction between  $k_{\min,2}$  and  $k_{\min}$  is needed because the original input files taper the slip-dependent bounds at depth while keeping the pressure-dependent floor constant. They are both  $10^{-19} \text{ m}^2$  above 30 km. The model excludes dilatancy, thermal pressurization, fluid exchange with the country rock, and changes in storage, as does the featured paper model.

For steady sliding, the reference permeability is

$$k_{\text{ss}}^* = \frac{k_{\text{max}} V_p / L + k_{\min} / T}{V_p / L + 1 / T}. \quad (8)$$

At depth in a region of constant properties, a constant effective stress requires  $k_{\infty} = \eta q_0 / (\partial_z \sigma - \rho g)$ . Consequently,

$$(\sigma - p)_{\infty} = -\sigma^* \log \left( \frac{k_{\infty} - k_{\min,2}}{k_{\text{ss}}^* - k_{\min,2}} \right). \quad (9)$$

This provides an independent scale for Figure 1: approximately 39.4 MPa in the featured case. The nonlinear permeability law makes pressure follow the normal-stress gradient after the shallow transition.

| Parameter                | Value                                                                         | Meaning                               |
|--------------------------|-------------------------------------------------------------------------------|---------------------------------------|
| $\mu, \eta_{\text{rad}}$ | 32.4 GPa, 4.68 MPa s m <sup>-1</sup>                                          | Elasticity and radiation damping      |
| $V_p, V_0, f_0$          | 10 <sup>-9</sup> m s <sup>-1</sup> , 10 <sup>-6</sup> m s <sup>-1</sup> , 0.6 | Friction reference values             |
| $d_c, L$                 | 0.002 m, 1 m                                                                  | State and permeability slip distances |
| $n, \beta, \eta$         | 0.01, 10 <sup>-9</sup> Pa <sup>-1</sup> , 10 <sup>-4</sup> Pa s               | Fluid storage and viscosity           |
| $\rho, g, \sigma^*$      | 1000 kg m <sup>-3</sup> , 9.8 m s <sup>-2</sup> , 30 MPa                      | Gravity and stress sensitivity        |
| $\partial_z \sigma$      | 22.05 MPa km <sup>-1</sup>                                                    | Original input normal-stress gradient |
| $T$                      | 10 <sup>7</sup> , 10 <sup>8</sup> , 10 <sup>9</sup> , 10 <sup>10</sup> s      | Healing times 0.317–317 yr            |
| $q_0$                    | $3 \times 10^{-9} \text{ m s}^{-1}$                                           | Except the 10 <sup>7</sup> s case     |

**Table 2:** Principal parameters. The exact piecewise-linear  $a(z)$  and  $b(z)$  profiles and depth-dependent permeability bounds are retained in the original input files. A year is 365 days.

### 3 Details recovered from the implementation

Several choices needed to run the model are more specific in the source inputs than in the article’s parameter table.

1. The input normal stress is 22.05 $z$  MPa for  $z$  in km. Friction uses  $\max(1 \text{ MPa}, \sigma - p)$ , while the permeability law uses the unmodified  $\sigma - p$ . Adding archived pressure to archived effective stress therefore does not recover the input normal stress where the frictional floor is active.
2. Both bounds in Equation (7) interpolate linearly between their shallow values at 30 km and a common constant at 60 km. That constant is  $10^{-17}$ ,  $9.1 \times 10^{-17}$ ,  $5 \times 10^{-16}$ , and  $9.091 \times 10^{-16} \text{ m}^2$  in the four source input files. The  $T = 10^{10} \text{ s}$  archive instead implies approximately  $9.09 \times 10^{-16} \text{ m}^2$  at depth, a small input/archive discrepancy.
3. The original files declare  $L_z = 500 \text{ km}$ , but their supplied grid and all archived depth vectors end at 578.952681 km. The new calculations use that actual extent and  $L_y = 500 \text{ km}$ . This choice can be changed at the command line.

4. The short-healing input uses  $q_0 = 3.3764 \times 10^{-10} \text{ m s}^{-1}$ ; the article reports  $3.3 \times 10^{-10} \text{ m s}^{-1}$ . We use the input value in new simulations and leave archived fields unchanged.
5. Archived permeability is in  $\text{km}^2$ , despite the SI units used in the published plots. The reconstruction converts it to  $\text{m}^2$  before plotting. Archived pressure and effective stress are in MPa, and depth is in km.

These details do not introduce new physical mechanisms. They specify the particular implementation that generated the available results and prevent silent differences in a new calculation.

## 4 Numerical verification and new integrations

The C++ code combines the spectral elastic operator with conservative hydraulic finite volumes. The hydraulic unknown is excess pressure  $h = p - \rho g z$ , so the gravitational flux cancels analytically. Face conductivities are arithmetic averages; the lower node has a half control volume and the prescribed incoming flux. Pressure stages use a nonlinear backward-Euler solve. At a midpoint stage this produces the implicit-midpoint pressure update for the full step. Step doubling estimates the temporal error in slip, state, reference permeability, and pressure. Long steps use Newton iteration with a cosine-transform elastic Jacobian and a preconditioned conjugate-gradient solve; steps shorter than 3000 s can use an explicit mechanical midpoint. A fully coupled Dormand–Prince 5(4) method is also available. Early integrations use it only below one second. The final solver permits it below 3000 s when  $\Delta t \max(k)/(n\beta\eta\Delta z^2) < 0.2$ , and checks a limit of 0.4 at every stage. Other stages retain implicit pressure. These switches change the numerical solution procedure, not the governing equations.

The final implementation optionally evaluates the independent nodal friction solves with OpenMP, using eight threads by default. The elastic transforms, hydraulic solves, and floating-point reductions retain their original order. A one-second seismic continuation from an identical checkpoint produces byte-identical field records, final states, and step histories with one and eight threads. Recorded wall times are 125 and 51 s, respectively. The late continuation of the longest-healing case uses this parallel version; its switch checkpoint is retained. This optimization changes execution time without changing the tested numerical trajectory.

The verification suite tests an elastic cosine mode and uniform loading, the frictional traction balance over slip rates  $10^{-20}$ – $10 \text{ m s}^{-1}$ , preservation of a discrete steady Darcy flux, an exact discrete diffusion eigenmode, and fluid mass balance. The largest pressure error in the steady test is about  $2 \times 10^{-6} \text{ Pa}$ ; the manufactured diffusion error is below  $10^{-9} \text{ Pa}$  and its integrated mass-balance error is below  $10^{-14} \text{ m}$  of fluid volume per area. A uniform spring-slider test gives a temporal error ratio of 4.003 when the time step is halved, consistent with second-order midpoint integration. The Dormand–Prince test approaches fifth order, with an error ratio of about 28.4 on the tested steps. Checkpoint continuation is also compared with uninterrupted integration. These tests verify individual operators and temporal order. They do not establish spatial convergence of a nonlinear earthquake sequence.

An integration from the identical 12-year checkpoint through year 14.7 tests the expanded fifth-order range against the earlier subsecond-only rule. Wall time decreases from 555 to 480 s in this recorded comparison. The first sampled seismic crossing changes by approximately 2 s and the effective-stress difference is 0.5 Pa RMS. This is smaller than the separate tolerance-refinement difference. The sampled peak changes from 1.04 to 1.00  $\text{m s}^{-1}$ , within the variation seen in the separate tolerance comparison. The coarser and finer featured-case runs and the  $T = 10^{10} \text{ s}$  run use the expanded range in later continuations; their switch states, times, and source revisions are retained with the results.

The reported integrations retain their source-version history in `results/run_provenance.json`. The final implementation sets imported surface pressure to exactly zero; the early integrations retained the tiny surface residual in the archived data, at most 1.904 Pa across the four cases. The continuation records identify which version was used. The repository therefore preserves the actual computational provenance rather than implying that all outputs were generated by the final source revision.

Reducing the featured-case time tolerance from  $10^{-4}$  to  $2.5 \times 10^{-5}$  over the first year changes effective stress by about 34 Pa RMS and  $\log_{10} V$  by about  $5.7 \times 10^{-4}$  decades RMS over 2–24.5 km depth. This comparison includes interpolation between saved output times. The temporal difference is smaller than the corresponding differences from the original archive and between spatial resolutions.

A separate temporal refinement starts from the identical 12-year checkpoint and integrates through the first earthquake. Reducing the tolerance by the same factor shifts the first  $1 \text{ mm s}^{-1}$  crossing approximately 169 s earlier (Figure 16). The sampled peak velocity changes from 1.04 to  $0.97 \text{ m s}^{-1}$ . The event-time shift is small compared with the tens of days separating the new and archived events, while the roughly 7% amplitude change precludes a claim of fully converged rupture waveforms.

An approximate restart provides a stricter comparison with the archived interseismic evolution than an arbitrarily phased fresh cycle. We recover  $k^*$  algebraically from pressure and permeability. To recover state, write  $\psi = f_0 + b \log(V_0 \theta / d_c)$ , for which  $\dot{\theta} = 1 - V \theta / d_c$ . Starting with  $\theta = 1 \text{ s}$  at the first archived sample, we integrate this law using each archived slip increment, assuming constant velocity over that interval. We then infer traction from the friction law and archived velocity. The restart is taken approximately four years into each archive. Changing the assumed initial  $\theta$  from 1 to  $2 \times 10^6 \text{ s}$  changes reconstructed  $\theta$  by at most about 1.6% at these restart times. This conditional sensitivity quantifies memory of the assumed initial state; it does not bound error from temporal sampling, interpolation, or an original initial state outside that tested range.

| $T$ (yr) | $\Delta z$ (m) | Duration (yr) | Stress RMS (MPa) | $\log_{10} k$ RMS | $\log_{10} V$ RMS |
|----------|----------------|---------------|------------------|-------------------|-------------------|
| 0.317    | 35.34          | 48.00         | 0.252            | 0.175             | 0.299             |
| 3.17     | 70.67          | 35.00         | 3.12             | 0.417             | 1.41              |
| 3.17     | 35.34          | 35.00         | 2.9              | 0.375             | 1.31              |
| 3.17     | 17.67          | 16.00         | 0.663            | 0.0621            | 0.301             |
| 31.7     | 35.34          | 80.00         | 1.94             | 0.145             | 0.631             |
| 317      | 35.34          | 180.00        | 0.899            | 0.0426            | 0.662             |

**Table 3:** New C++ runs compared with the archived fields over 2–24.5 km depth and the available overlap in time. Duration gives the full new integration; the  $T = 0.317 \text{ yr}$  comparison ends 44.82 yr after its restart because the archive ends there. Logarithmic errors are in decades, with velocities below  $10^{-30} \text{ m s}^{-1}$  set to that value for this diagnostic only. Each run starts near four years into its archive. RMS values use 101 uniformly spaced physical times over the overlap and 226 depths at 100 m spacing; this sampling does not resolve every seismic peak.

The finer featured-case run covers 16.00 years after the approximate restart. Its effective-stress RMS difference from the archive is 0.663 MPa, and its logarithmic slip-rate RMS difference is 0.301 decades. These are comparisons with a reconstructed initial condition, rather than errors relative to an exact solution. Over the first 12.00 years before the earthquake, the two finest featured-case meshes differ by 0.00832 MPa RMS in effective stress. This smooth-field comparison does not establish convergence of the later seismic sequence.

The first featured-case seismic crossing occurs 37.97 and 39.11 days earlier than in the archive on the 35.34 and 17.67 m meshes, respectively. The 1.13-day change between these two meshes is smaller than the 19.82-day change between the coarser pair. Spatial refinement therefore reduces their mutual

| $\Delta z$ (m) | First crossing (archive-relative yr) | Difference from archive (days) |
|----------------|--------------------------------------|--------------------------------|
| 70.67          | 18.654188                            | -18.15                         |
| 35.34          | 18.599886                            | -37.97                         |
| 17.67          | 18.596783                            | -39.11                         |

**Table 4:** First sampled crossing of  $1 \text{ mm s}^{-1}$  between 2 and 25 km depth in the featured-case spatial refinements. Times retain the archive’s origin; its corresponding crossing occurs at 18.703924 yr. A negative difference denotes an earlier event. The comparison includes uncertainty in the reconstructed initial state and does not measure error relative to an exact solution.

timing difference but leaves an offset from the archive. The tested temporal changes are much smaller than this offset. These comparisons motivate further checks of the reconstructed state and original discretization; they do not establish convergence of the full swarm sequence.

Before the first earthquake, the new solutions retain the upward advance of the slipping region (Figure 14). The later featured-case calculation also generates repeated shallow slip bursts, but their timings and depth extents differ from the deposited sequence (Figure 13). The effective-stress, permeability, and flux maps show the accompanying hydraulic front (Figure 17). Thus the new integration recovers the coupled feedback qualitatively while its later trajectory separates from the archive. A shifted sharp front can produce a substantial pointwise field difference, so the tabulated RMS values should be read together with these maps.

The short-healing case generates repeated ascending aseismic pulses during the interval before its earthquake (Figure 15). The two longer-healing cases also exhibit upward slip-front evolution. Their intermediate bursts and large ruptures occur at different times from the archive. These calculations test all four published healing-time regimes. Spatial refinement was performed only for the featured case; the other three cases use one resolution. Their recurrence times and event amplitudes are therefore not established as independently converged predictions.

## 5 Figure reconstruction and interpretation

Figure 1 recovers the steady effective-stress plateau and the permeability laws by direct calculation. The archived results in Figures 2–5 show the intended physical feedback: sealing reduces fluid escape, the growing pressure perturbation weakens the fault, and slip permits further upward transport. The swarm sequence is present in the supplied data and is visible in the enlarged time window. Its recovery here is a successful data and plotting reproduction.

For Figure 6, we define the advancing front as the shallowest  $V = 10^{-10} \text{ m s}^{-1}$  crossing below 2 km. The displayed 25-year windows start at 12, 6, 20, and 25 years into the four archives, respectively. They exclude the initial postseismic retreat and are explicitly chosen plotting windows; the article does not state its corresponding offsets or threshold. Linear fits over declared advancing intervals give approximately 0.409, 0.126, and 0.033  $\text{km yr}^{-1}$  for  $T = 3.17, 31.7,$  and 317 yr. These are close to the published scales of 0.38, 0.12, and 0.03  $\text{km yr}^{-1}$ , but do not reproduce the exact graphical construction. The short-healing case has repeated pulses, so its window-averaged migration rate is not a measure of the much faster individual pulse fronts labelled in the published figure.

All reconstructed maps retain every saved time sample and use every fifth depth node, giving 50 m spacing above 20 km. The archive itself resolves 10 m there. The sample-index axes expand short seismic intervals but are not physical time axes. Cumulative slip is referenced separately at each depth to its value at the beginning of the plotted window. Interseismic profiles are sampled at the

captioned intervals. Red coseismic profiles are interpolated at one-second intervals within archived bursts with maximum velocity exceeding  $10^{-3} \text{ m s}^{-1}$ . These conventions are reproducible but need not match every contour or annotation in the original plotting scripts. Cumulative-slip panels use the longer deposited sequence, while velocity maps emphasize a representative earlier interval, following the published comparisons.

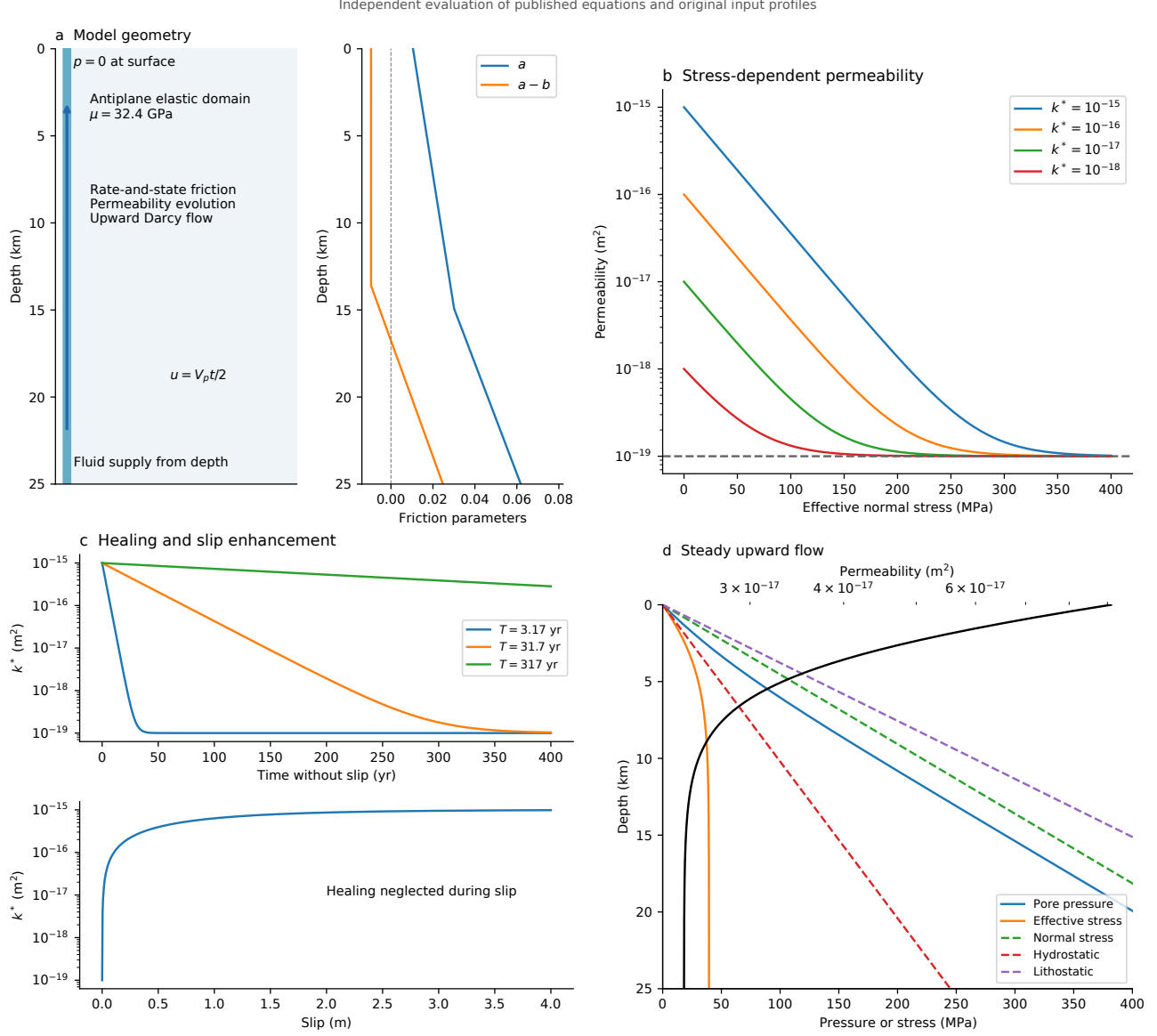
## 6 What remains unreproduced

The constitutive and steady-flow results are independently recovered, and the deposited transient fields recover the principal published patterns. An exact, independent, resolution-converged reproduction of the complete earthquake cycles, individual swarm events, and slow-slip recurrence is not established in this exercise. The archive lacks exact restart state and traction, and the new code uses a spectral elastic operator and second-order hydraulic volumes instead of the original fourth-order summation-by-parts discretization. Agreement between archived figures cannot resolve those differences. The fixed-pressure reference histories in Figures 2, 9, and 11 were reconstructed from the archive. Although the new solver supports fixed pressure, the new sequence integrations reported here examine the coupled cases.

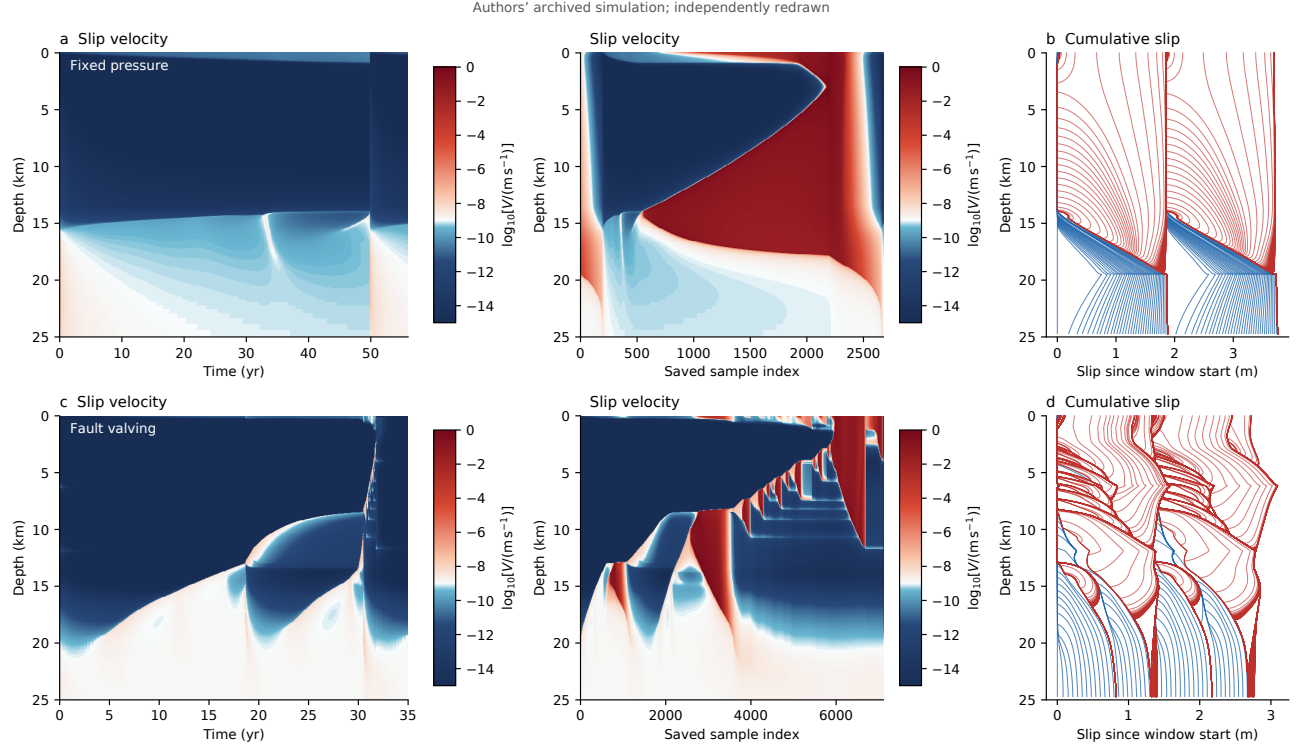
The two supplementary movies were not recreated. No claim is made that the plotted source trajectories were generated by the new C++ code. The bounded result is a transparent implementation, a recoverable set of published figures, and quantitative tests that expose the remaining initialization and numerical comparisons. A complete sequence benchmark requires saved original states and stresses, followed by spatial and temporal convergence tests across both interseismic fronts and coseismic events.

## References

- [1] Zhu, W., Allison, K. L., Dunham, E. M., and Yang, Y. (2020). Fault valving and pore pressure evolution in simulations of earthquake sequences and aseismic slip. *Nature Communications*, 11, 4833. [doi:10.1038/s41467-020-18598-z](https://doi.org/10.1038/s41467-020-18598-z).
- [2] Zhu et al. (2020). Simulation data and source-data information. Open Science Framework. [doi:10.17605/OSF.IO/9YGRP](https://doi.org/10.17605/OSF.IO/9YGRP).
- [3] Scycle source code, branch Zhu\_et\_al\_2020, commit d65e248900e193add0cc901ee7ee4d8ee7a2d50a. [https://bitbucket.org/kallison/scycle/src/Zhu\\_et\\_al\\_2020/](https://bitbucket.org/kallison/scycle/src/Zhu_et_al_2020/).

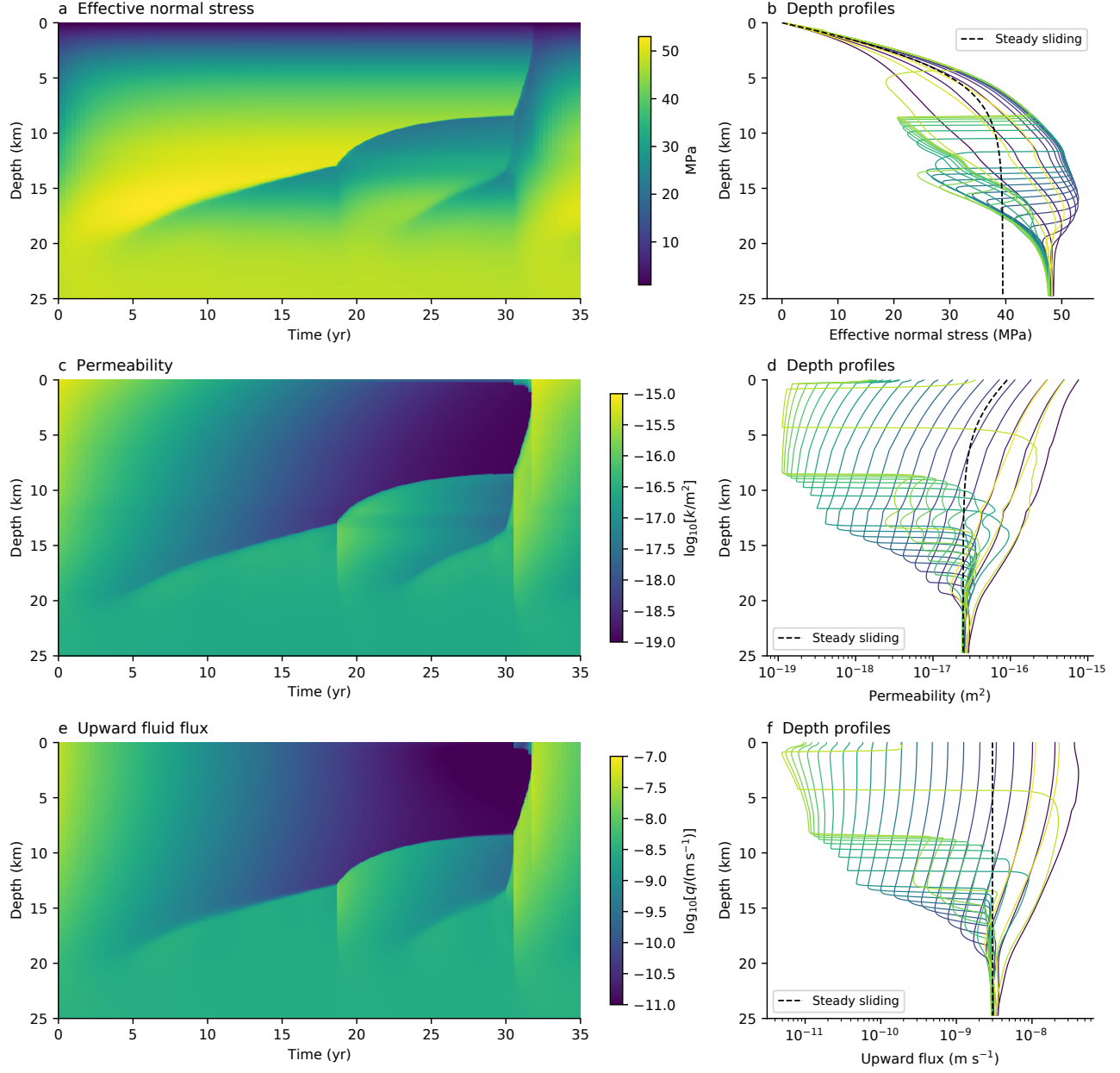


**Figure 1:** Independent reconstruction of published Figure 1. (a) Schematic of the upper fault and the original piecewise-linear friction profiles; the prescribed basal flux lies at the full computational depth, 578.95 km. (b) Pressure-dependent permeability for four reference permeabilities. (c) Healing without slip and enhancement during slip, with the competing process omitted in each limiting calculation. (d) Pressure, normal stress, effective stress, and permeability under steady upward flux. The featured parameters are  $T = 10^8$  s,  $q_0 = 3 \times 10^{-9}$  m s $^{-1}$ , and  $\sigma^* = 30$  MPa.



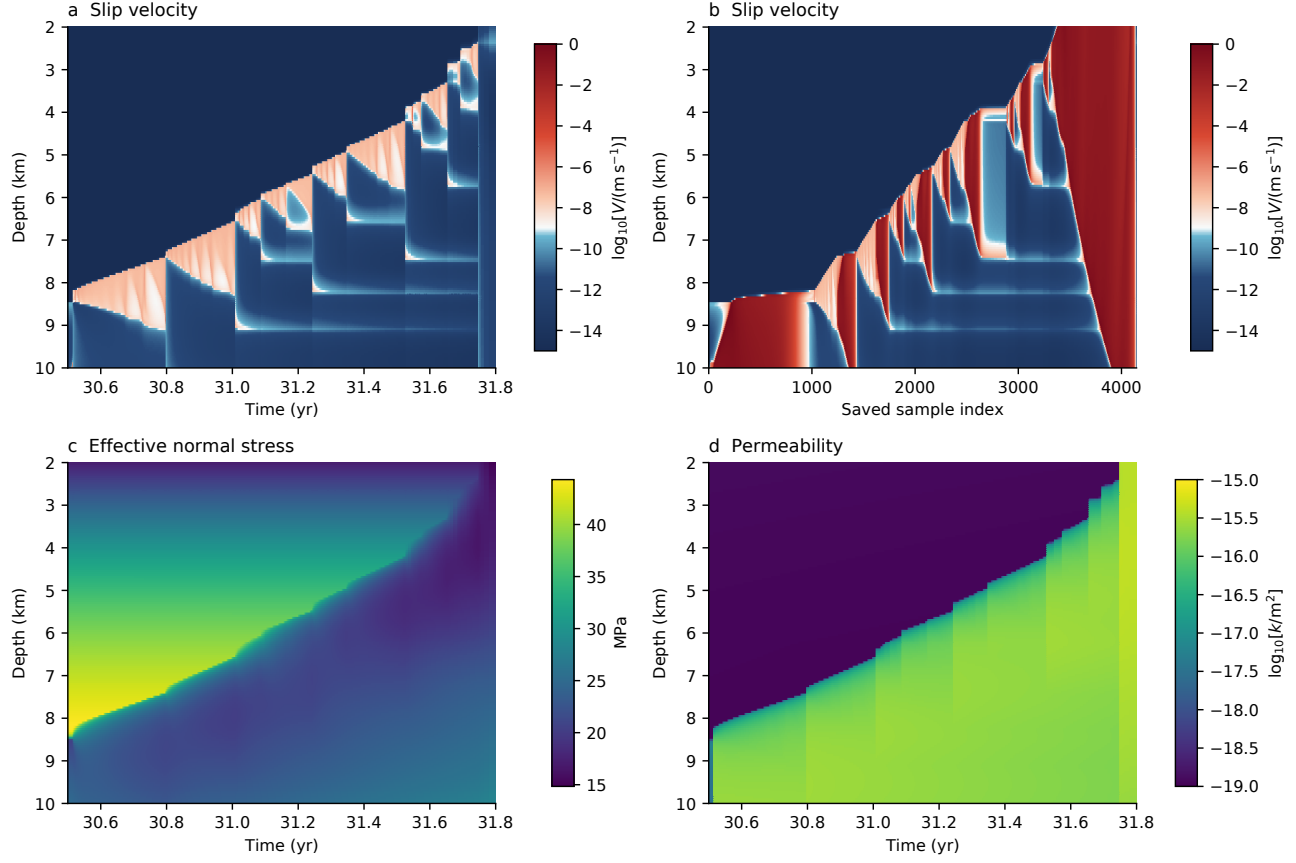
**Figure 2:** Reconstruction of published Figure 2 from the authors' archived  $T = 10^8$  s simulations. The upper row holds pore pressure fixed; the lower row includes fault valving. Velocity is plotted against physical time and saved sample index. Right panels show cumulative slip relative to each depth's initial value; blue curves are separated by 1.5 yr, and red curves represent one-second seismic snapshots. The advancing aseismic fronts and swarm sequence are recovered from the archive. These panels are not output from the new solver.

Authors' archived simulation; independently redrawn



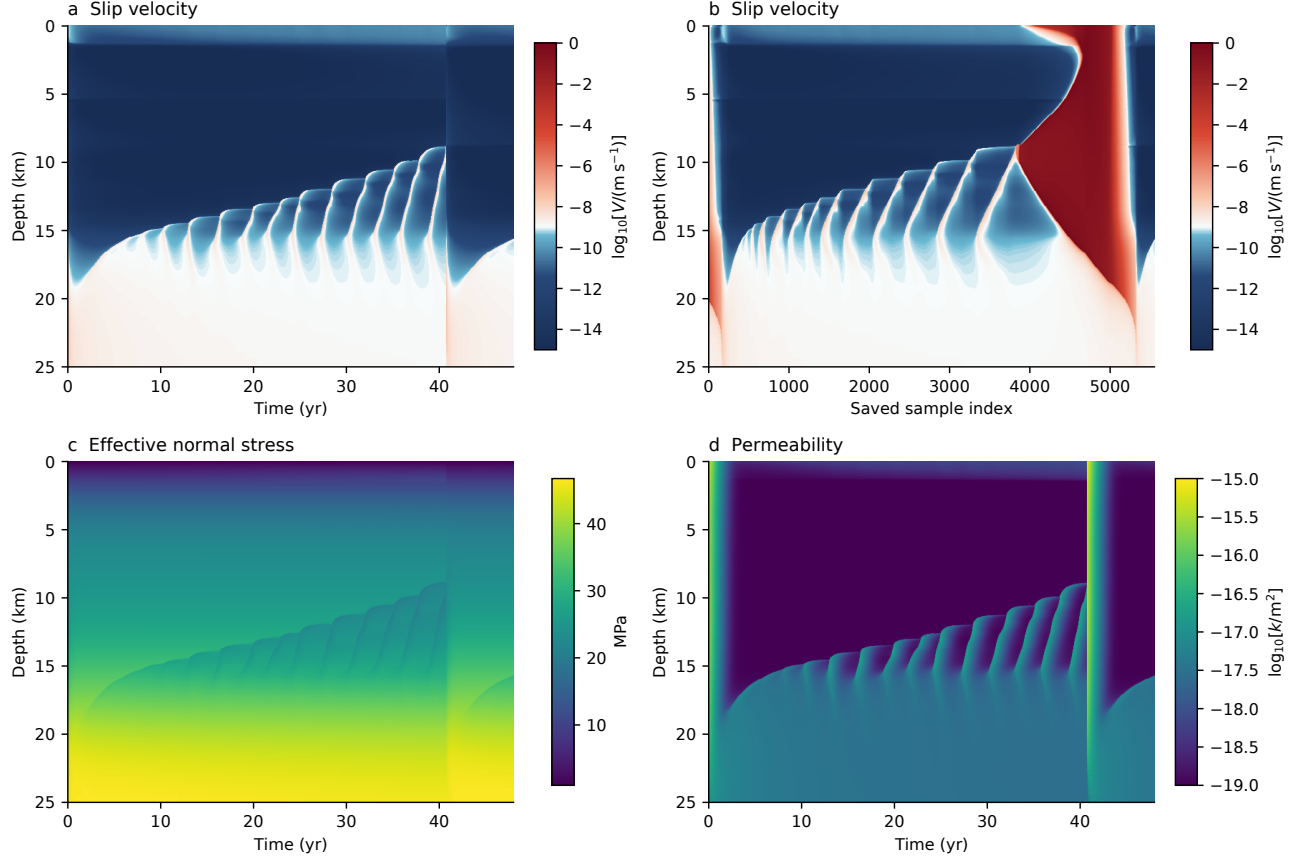
**Figure 3:** Reconstruction of published Figure 3 from the archived  $T = 10^8$  s fault-valving case. Left panels show effective stress, permeability, and upward flux. Right panels show profiles every 1.5 yr, with color progressing from purple at early times to yellow at late times; dashed black curves are independently calculated steady-sliding profiles. Nonpositive flux, if present, is masked on logarithmic maps. Profile colors and spacing differ from the original phase-based presentation.

Authors' archived simulation; independently redrawn



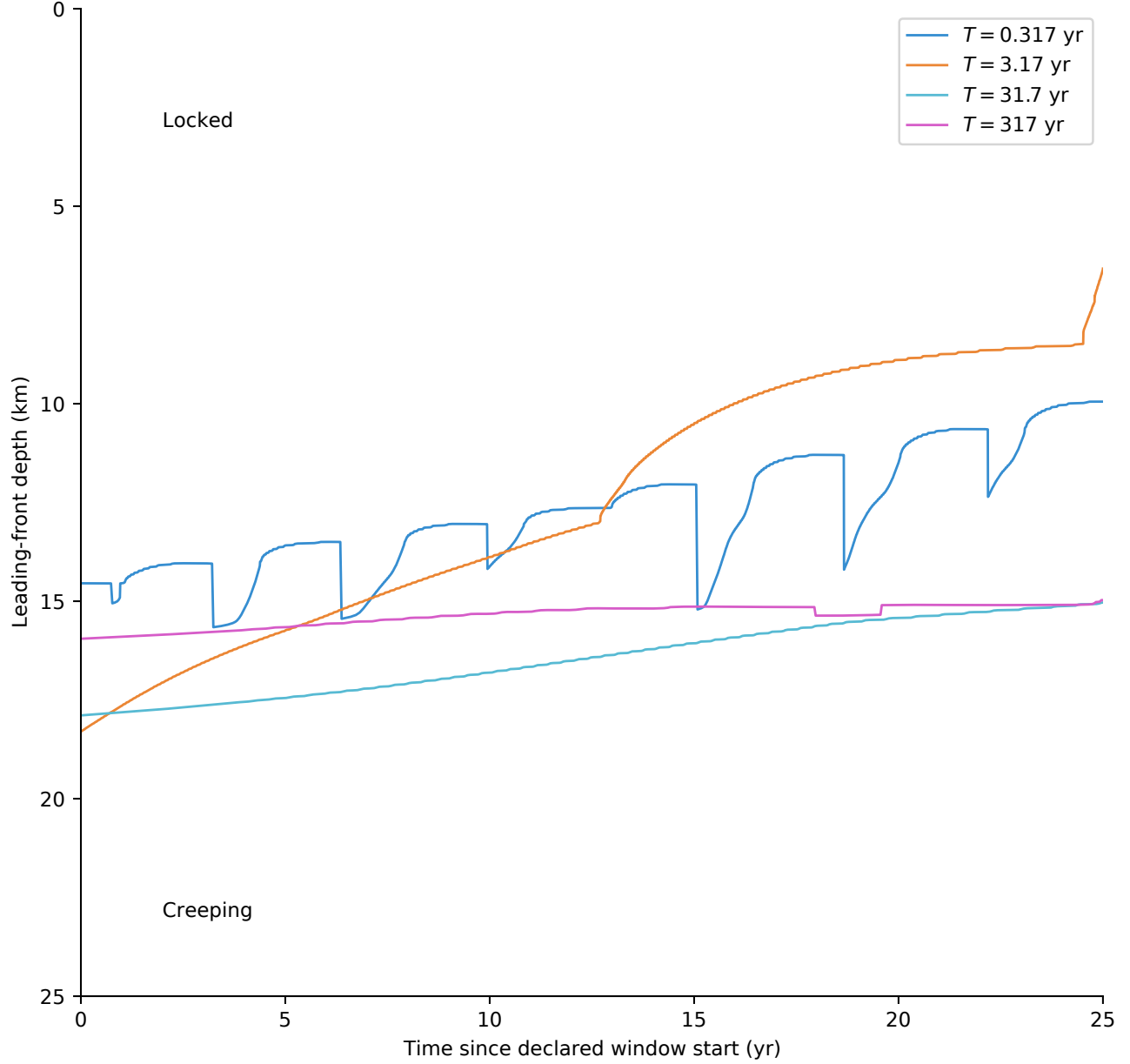
**Figure 4:** Reconstruction of published Figure 4 from the archived swarm interval, 30.5–31.8 yr after the first saved sample and 2–10 km depth. The upper panels show slip velocity against physical time and saved sample index. Lower panels show effective stress and permeability. The ascending pressurization front accompanies repeated slip bursts. The calculation recovers the archived sequence; it does not independently generate these events.

Authors' archived simulation; independently redrawn

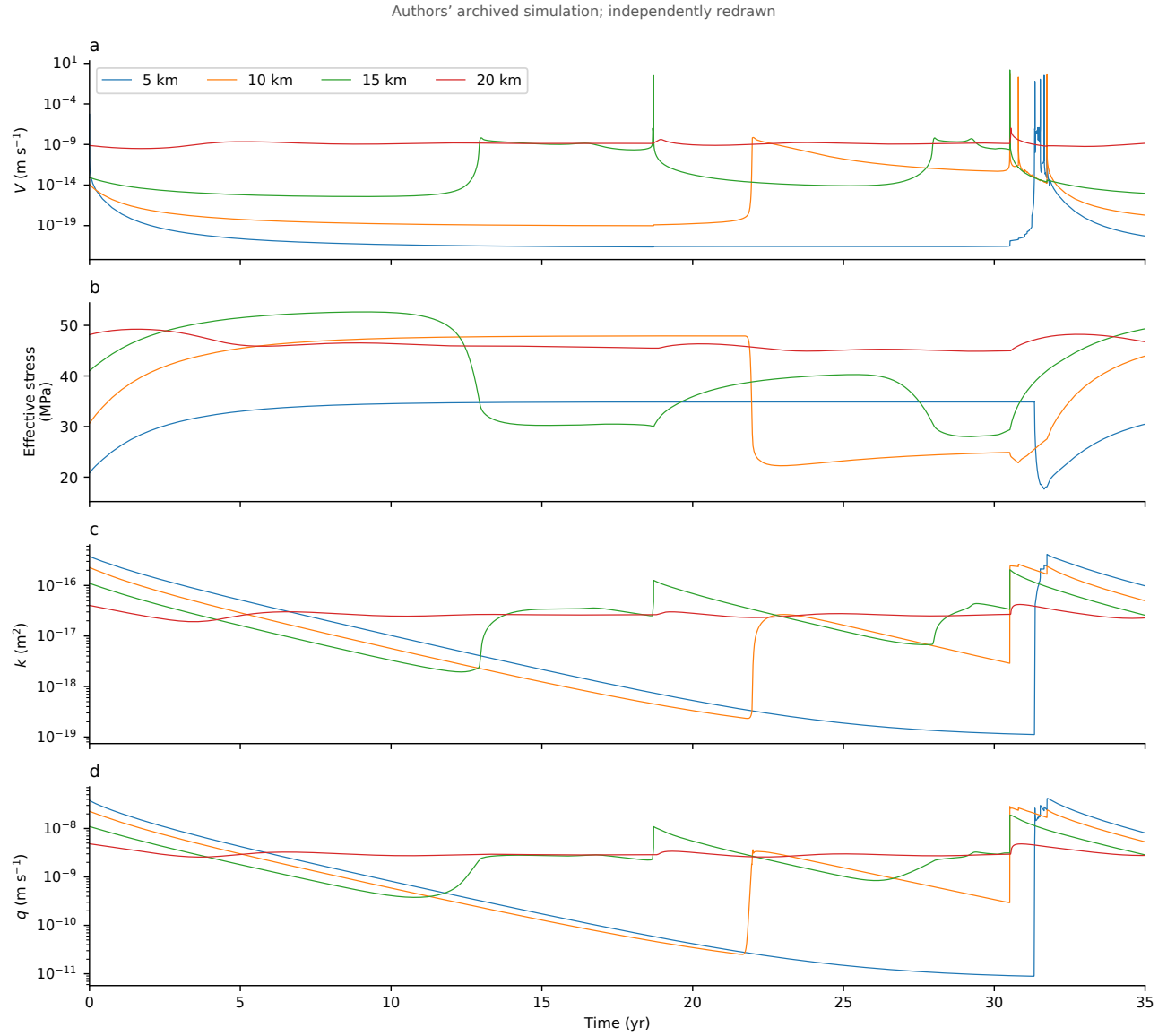


**Figure 5:** Reconstruction of published Figure 5 from the archived  $T = 10^7$  s case. The panels show velocity against time and sample index, effective stress, and permeability. Repeated aseismic pulses ascend from the base of the locked region. The source input flux is  $3.3764 \times 10^{-10} \text{ m s}^{-1}$ , slightly larger than the rounded value in the article caption.

Authors' archived simulation; independently redrawn

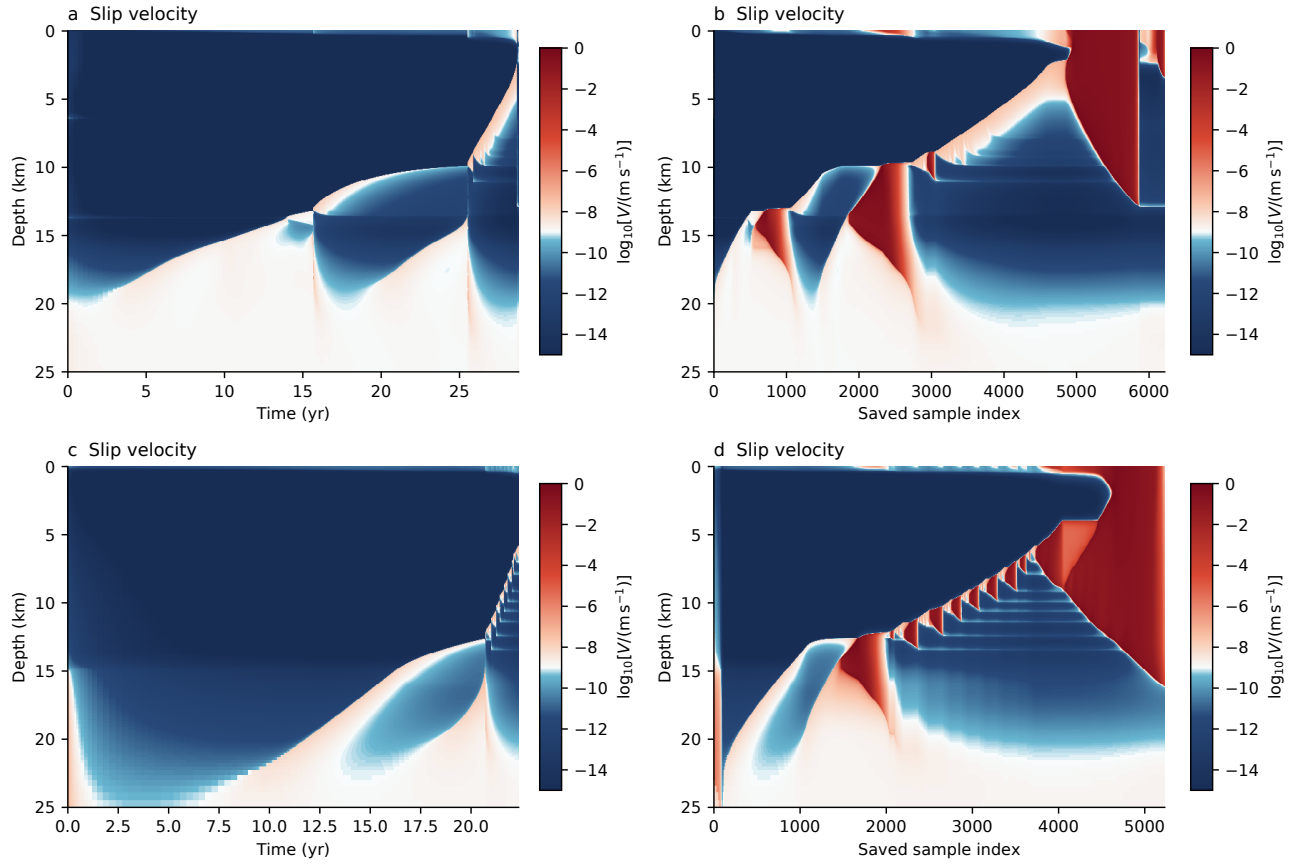


**Figure 6:** Reconstruction of the comparison in published Figure 6 using an operational front definition. Curves mark the shallowest  $V = 10^{-10} \text{ m s}^{-1}$  crossing below 2 km in the archived simulations. Window starts are 12, 6, 20, and 25 yr for increasing  $T$ . Longer healing times yield slower advancing fronts. The threshold and offsets are stated choices because the published graphical definition is not supplied. The short-healing curve includes repeated advance and retreat; its mean slope is not an individual pulse speed.



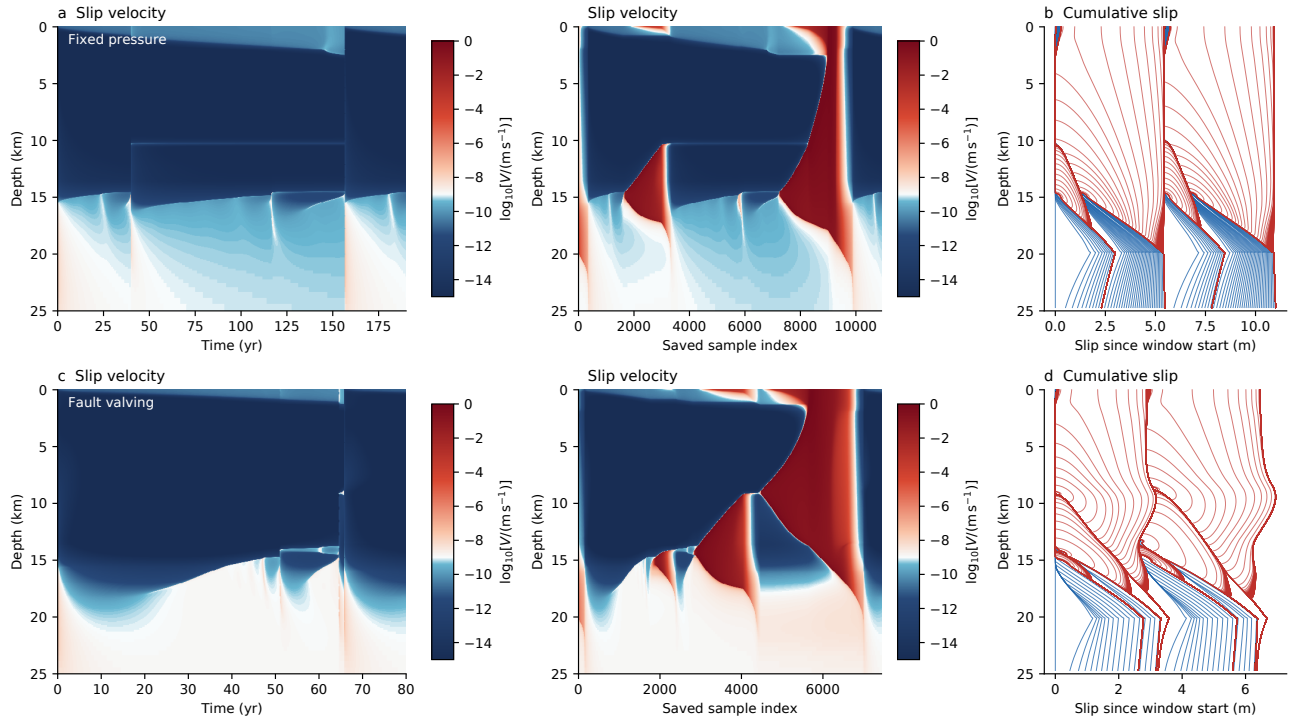
**Figure 7:** Reconstruction of Supplementary Figure 1. Archived velocity, effective stress, permeability, and upward fluid flux at depths nearest 5, 10, 15, and 20 km in the featured  $T = 10^8$  s case. Colors identify depth. The histories show pressure and permeability changing together over a cycle.

Authors' archived simulation; independently redrawn

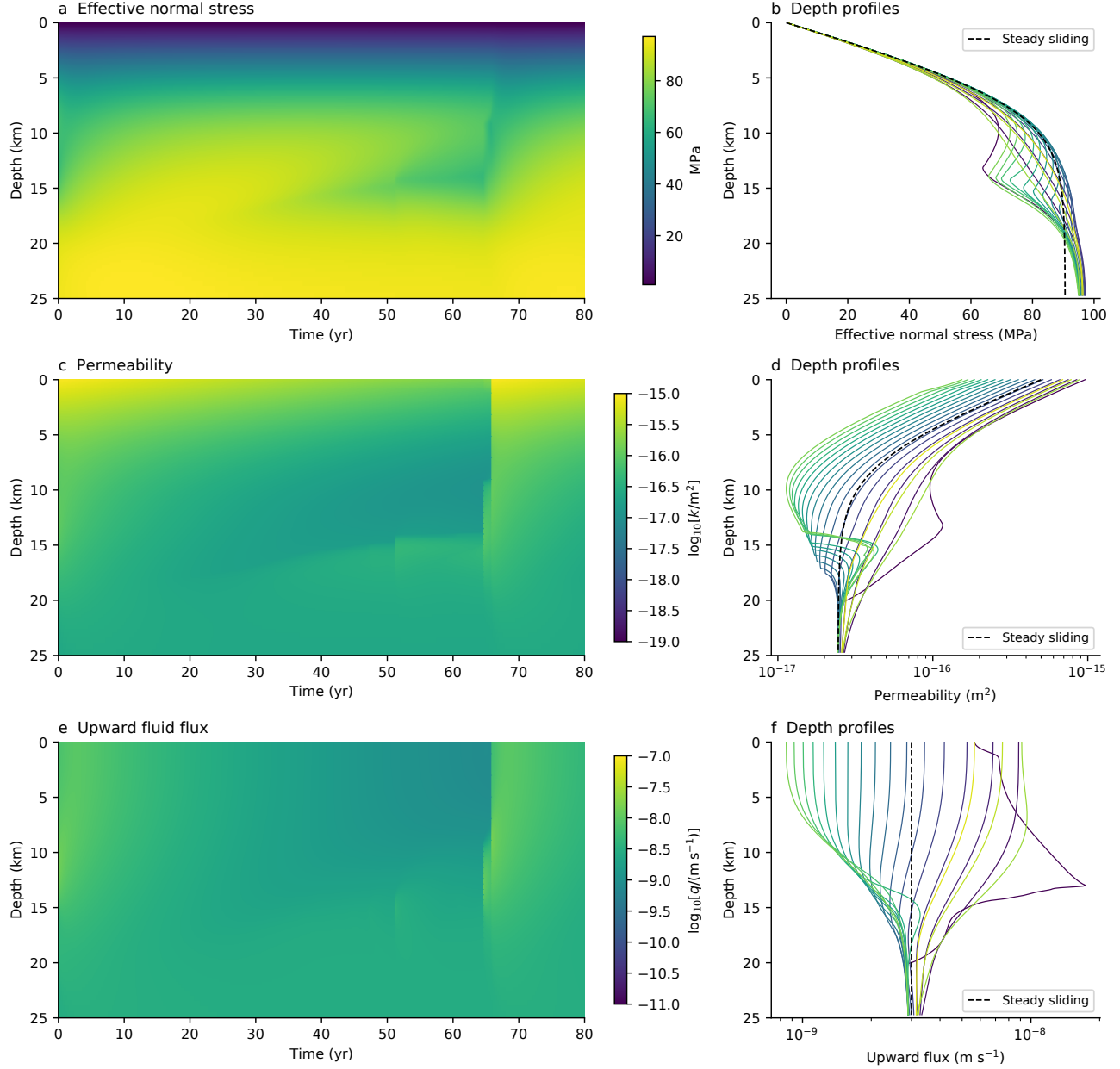


**Figure 8:** Reconstruction of Supplementary Figure 2 using the two alternate-cycle archives, `T1e8-sup1.mat` and `T1e8-sup2.mat`. Each row shows velocity against physical time and saved sample index. Their durations are those of the deposited files, approximately 28.8 and 22.4 yr. The differences between cycles indicate that the featured sequence is not exactly periodic.

Authors' archived simulation; independently redrawn

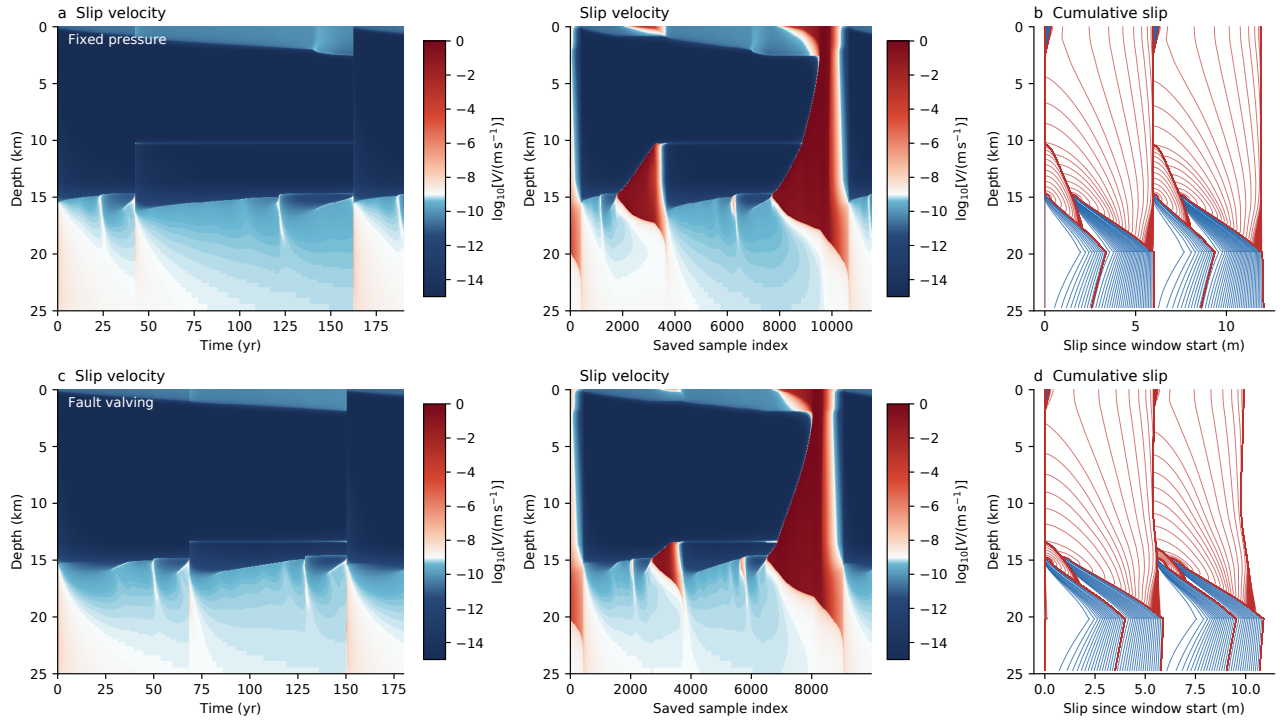


**Figure 9:** Reconstruction of Supplementary Figure 3 for  $T = 10^9$  s (31.7 yr). Fixed-pressure and fault-valving velocity maps cover 190 and 80 yr, respectively. Cumulative-slip profiles use the full deposited intervals, approximately 318 and 134 yr. Blue curves are separated by 4 yr, and red curves represent one-second seismic snapshots.



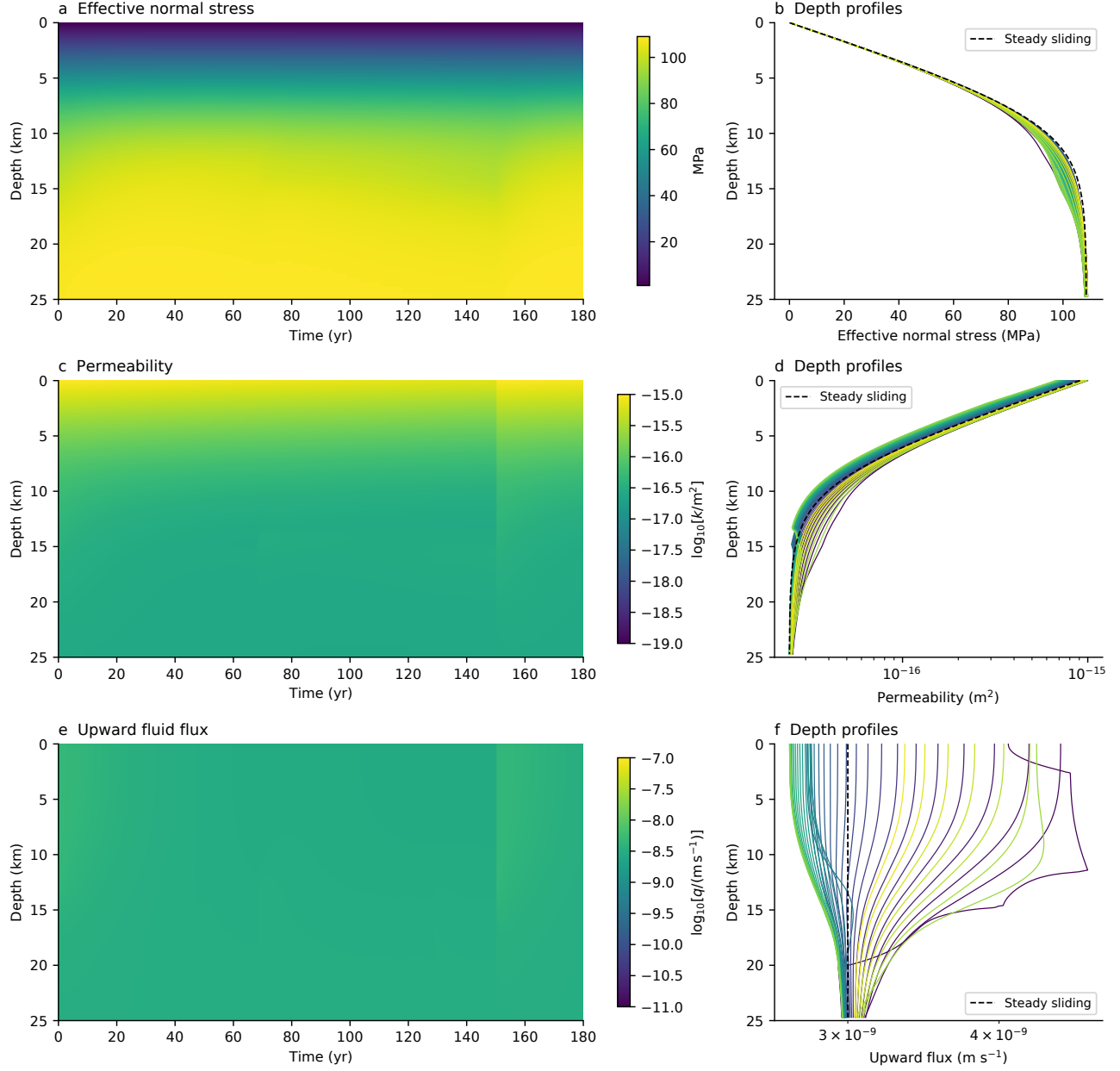
**Figure 10:** Reconstruction of Supplementary Figure 4 for  $T = 10^9$  s. Effective stress, permeability, and upward flux are shown as time–depth maps and profiles separated by 4 yr. Purple-to-yellow color indicates increasing time; black dashed profiles are independent steady-sliding solutions. Healing is slower than in the featured case, reducing the strength of the valving cycle.

Authors' archived simulation; independently redrawn

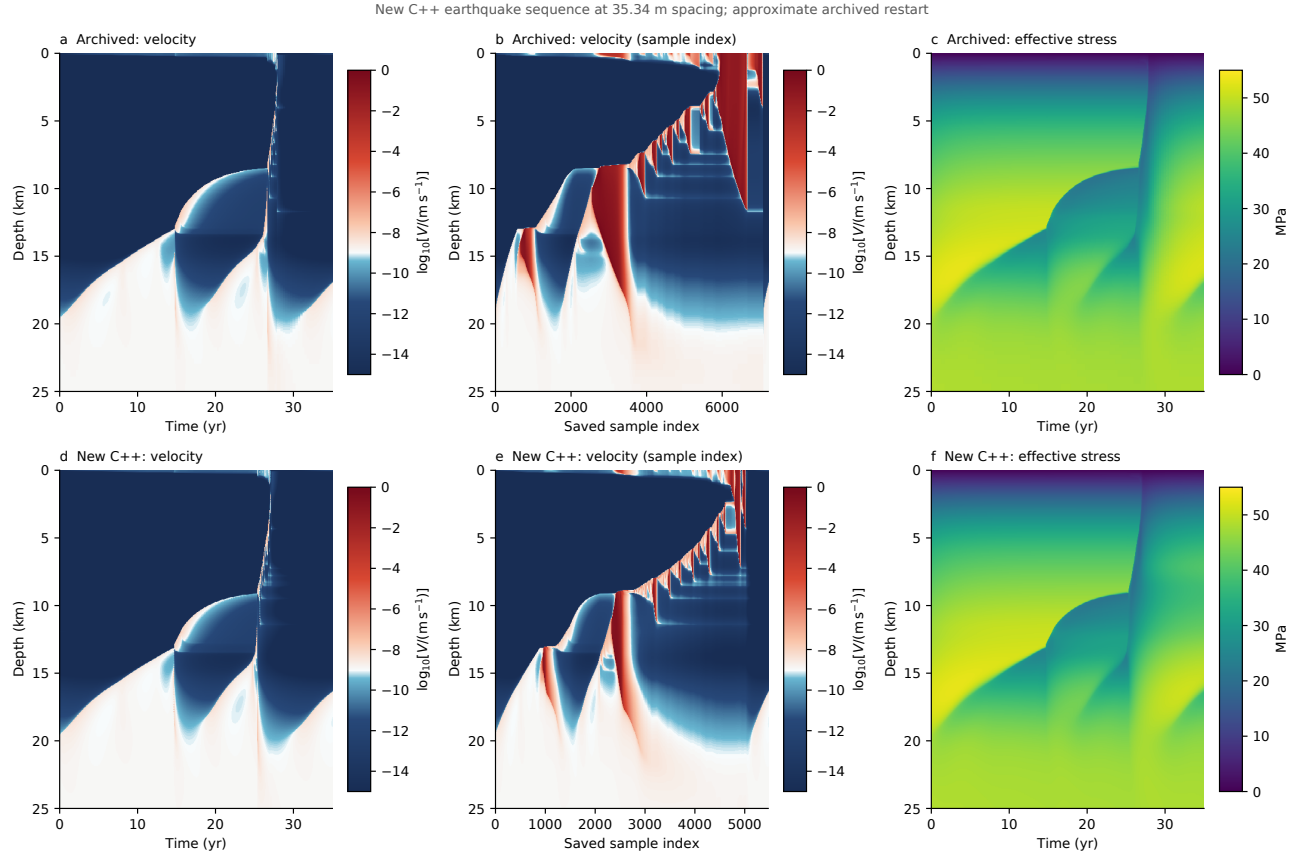


**Figure 11:** Reconstruction of Supplementary Figure 5 for  $T = 10^{10}$  s (317 yr). Fixed-pressure and coupled velocity maps cover 190 and 180 yr, respectively. Slip panels use the full deposited intervals, approximately 330 and 280 yr. Blue curves are separated by 4 yr, and red curves represent one-second seismic snapshots.

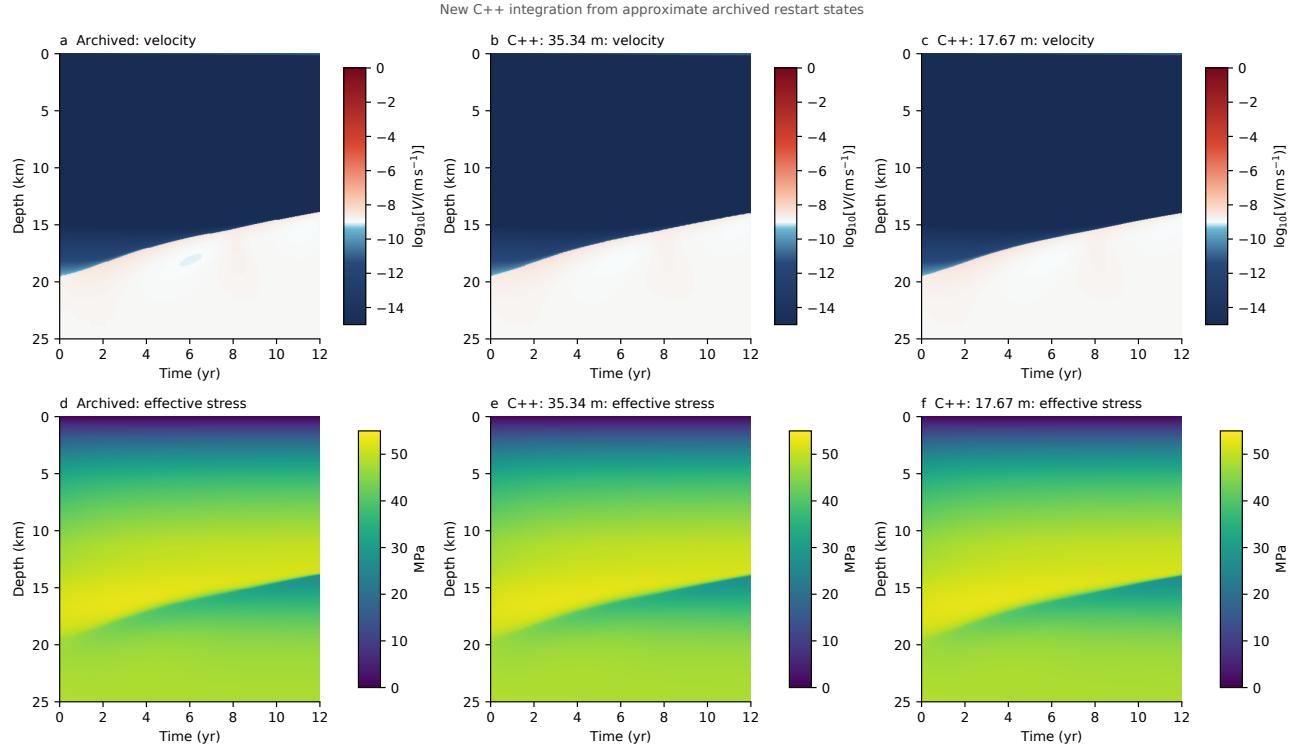
Authors' archived simulation; independently redrawn



**Figure 12:** Reconstruction of Supplementary Figure 6 for  $T = 10^{10}$  s. Effective stress, permeability, and upward flux are shown with profiles every 4 yr. The long healing time keeps the fault relatively permeable through much of the cycle. The archived weak pressure cycling is recovered, while this panel makes no claim about a new independently converged earthquake sequence.

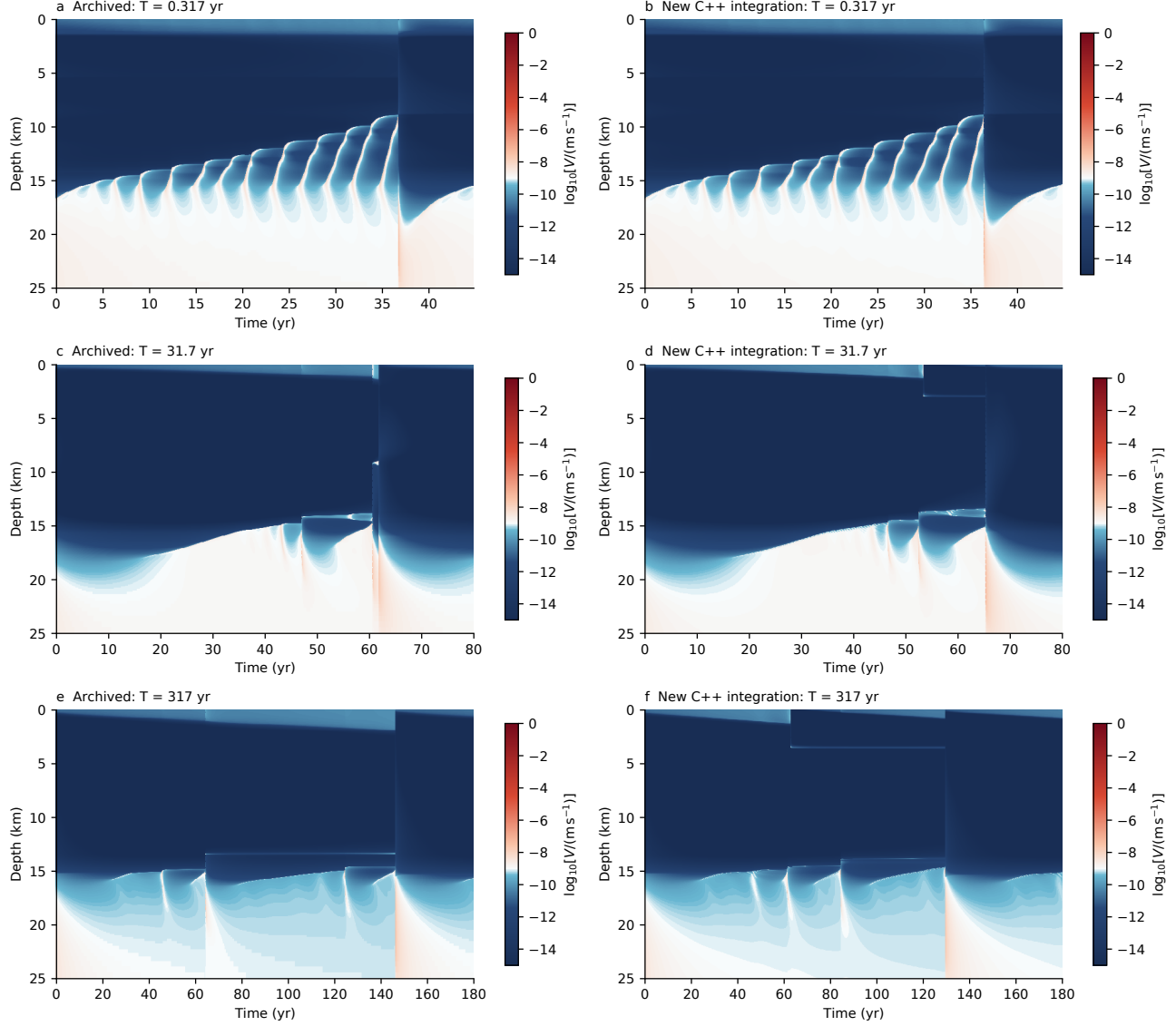


**Figure 13:** Independent C++ continuation of the featured fault-valving model, compared with the authors' archive over the same time interval. The upper row is archived and the lower row is newly integrated at 35.34 m spacing, starting from the reconstructed state about four years into the archive. Columns show velocity against physical time, velocity against saved sample index, and effective stress. The index panels expand rapid events; their spacing is not a measure of physical duration. These maps test the sequence of pressurization, aseismic migration, and seismic slip.

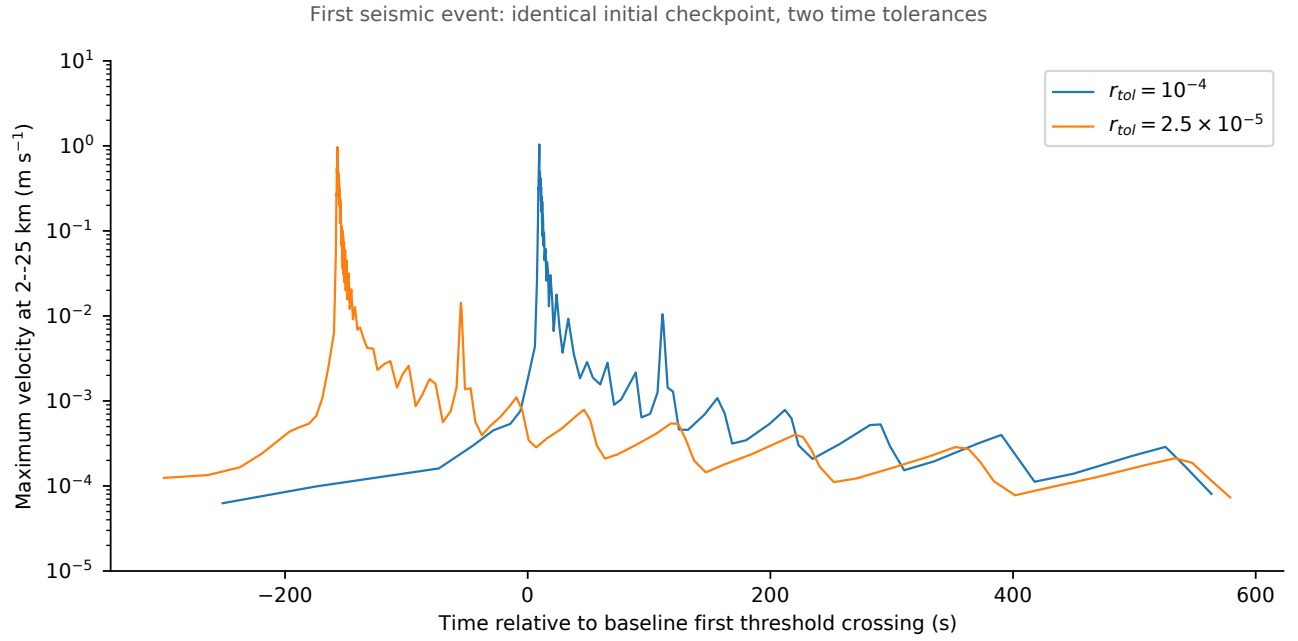


**Figure 14:** New C++ integrations compared with the authors’ archived featured case over the first 12 years after the approximate restart, before the first earthquake. The restart lies about four years into the archive. Columns distinguish archived fields from the new solutions. The comparison tests interseismic front evolution at 35.34 and 17.67 m spacing, subject to the initialization and discretization limitations discussed in the text.

Independent coupled integrations at other healing times; 35.34 m spacing

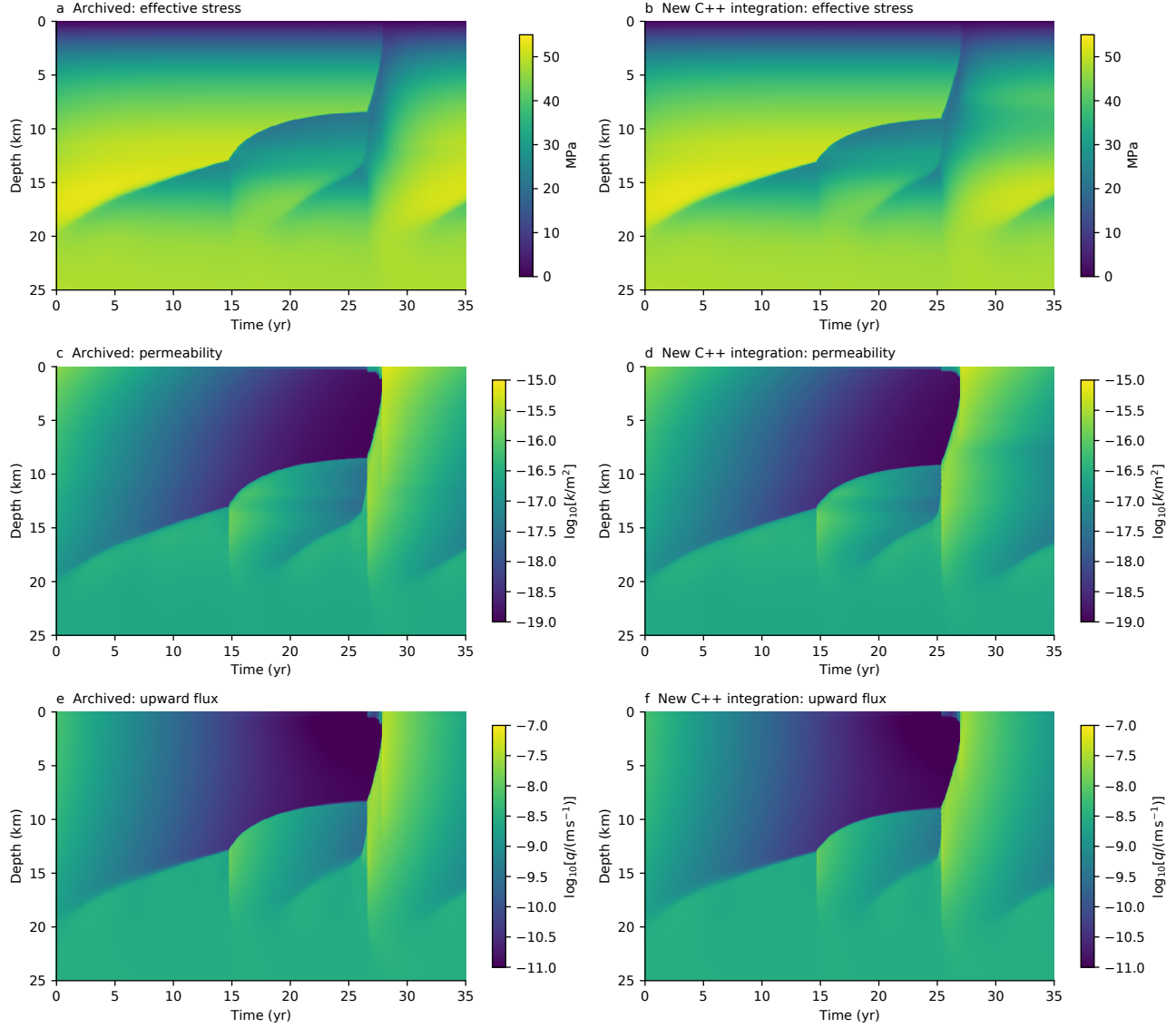


**Figure 15:** New C++ integrations for healing times of 0.317, 31.7, and 317 yr, compared with the corresponding archived velocity fields. Each row uses the same time interval in the archive and the new calculation, measured from its approximate restart about four years into the archive. The new runs use 35.34 m grid spacing. Differences in pulse propagation and seismic timing are direct tests of the independent model implementation, subject to initialization and discretization uncertainty.



**Figure 16:** Temporal refinement through the first featured-case earthquake at 35.34 m spacing. Both calculations start from the identical state saved at year 12 after the approximate restart. The curves show the maximum velocity between 2 and 25 km depth, with a common time origin at the baseline run’s first crossing of  $1 \text{ mm s}^{-1}$ . Their relative timing is retained. This comparison isolates the time-tolerance effect for this event and grid; it does not establish convergence of the later swarm.

Independent C++ hydraulic evolution at 35.34 m spacing



**Figure 17:** Hydraulic evolution in the new C++ featured-case calculation at 35.34 m spacing, compared with the corresponding archived interval. Rows show effective stress, permeability, and upward fluid flux. Permeability and flux color scales are logarithmic; nonpositive flux is masked. The new flux is evaluated on control-volume faces and plotted at its own depth coordinates. This comparison tests pressure and permeability changes accompanying the newly generated slip sequence.